Problem 2: Device. Feature System.

- Device → device Name
- Camera → inherits. Device.
- Music Player → inherit Device.
- Smartdevice → inherit both.

Requirement :- • store device name.
• Display it using child class.

Example

```
#include <iostream>
using namespace std;

class A {
  public:
    int x = 10;
};

class B: virtual public A {};
class C: virtual public A {};

class D: public B, public C {};

int main() {
    D obj;

    cout << obj.x; // work fine
}
```

Constructor order

A → B → C → D

Base class (A) is created only ones.

Problem statement 1: Student Intern System.

Create a system where:

- Person → name.
- Student → Inherit Person
- Employe → //
- Intern → Inherit Both Student and Em

Use virtual keyword to avoid duplicate Person. input

Requirement:- • name display → name

Problem statement 1: Teacher vs Student Identity

- Teacher → role();
- Student → role()
- Person → inherit Both.

Problem Statement 2: ~~Dat~~ Salary system.

- full Time → salary()
- Part Time → salary()
- Employee → Both. inherit.

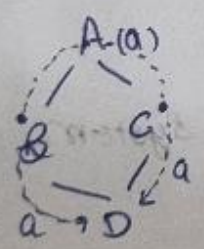
Problem statement 3: Device Control system.

- Camera → function start()
- Music Player → start()
- smartDevice → inherit Both.

Day 84 (Virtual Base class)

A virtual base class ensure that only one copy of a base class is inherited, Even if multiple path exist.

Why do we need it



without virtual

→ D got 2 copy of A

→ Same as ambiguity

with virtual

→ D got only 1 copy of A

Syntax

Class B: virtual public A{};

Class C: virtual public A{};